Module: General electronics

Tutorial Worksheet No. 2: **Passive filters**

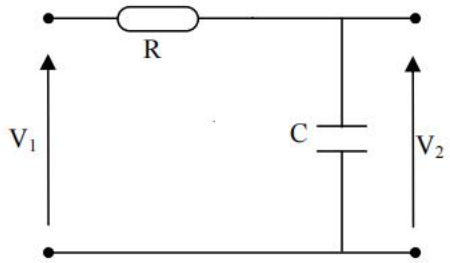
Duration: 3 weeks

Exercises to do in class: Week1: 1,2 Week2: 5, 7 Week3: 10

Assignment to submit: Week1: 3 Week2: 4 Week3: 8,9

Exercise 01 (Week 1)

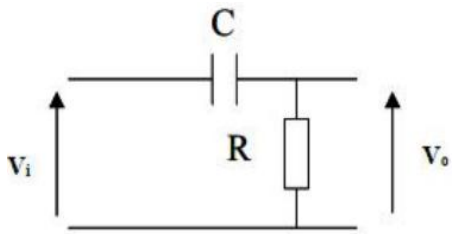
Given: R = 1kΩ; C = 1µF; Transfer function T(ω) = V_2 / V_1.



1. Determine the nature of the filter using simple qualitative reasoning.
2. Calculate the transfer function T(ω) and write it in its standard form.
3. Sketch the Bode plot for this filter.

Exercise 02 (Week 1)

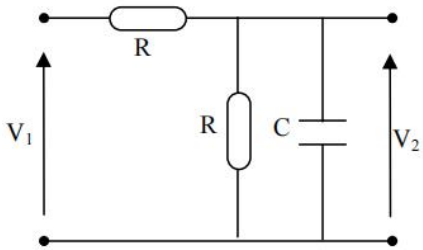
Given: R = 1kΩ; C = 1µF; Transfer function T(ω) = V_2 / V_1.



1. Determine the nature of the filter using simple qualitative reasoning.
2. Calculate the transfer function T(ω) and write it in its standard form.
3. Sketch the Bode plot for this filter.

Exercise 03

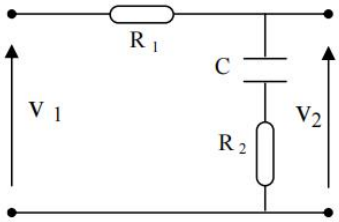
Given: R = 1kΩ; C = 1µF; Transfer function T(ω) = V_2 / V_1.



1. Calculate the transfer function T(ω) and write it in its standard form.
2. Sketch the Bode plot for this filter.

Exercise 04

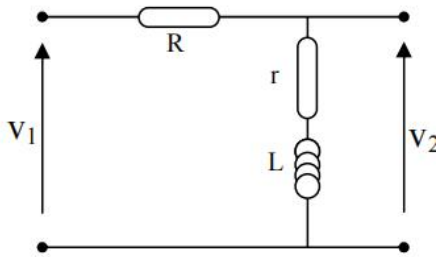
Given: R1 = 9kΩ; R2=1kΩ; C = 1µF; Transfer function T(ω) = V_2 / V_1.



1. Calculate the transfer function T(ω) and write it in its standard form.
2. Sketch the Bode plot for this filter.

Exercise 05 (Week 2)

Given: R = 9kΩ; r =1kΩ, L=100mH; Transfer function T(ω) = V_2 / V_1.



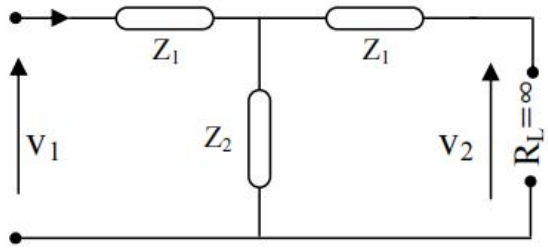
1. Calculate the transfer matrix of the two-port network (quadripole) and derive the transfer function T(ω)
2. Express it in the following form $$T = K_0 \frac{1 + j\frac{\omega}{\omega_1}}{1 + j\frac{\omega}{\omega_2}}$$

Provide the expressions and numerical values for K_0, ω_1, and ω_2.

3. Represent the response curves graphically on semi-logarithmic paper (Magnitude in dB and Phase).

Exercise 06

1) Consider the symmetrical T-network (two-port network) shown in the figure above. Express the elements of the admittance matrix in terms of the impedances Z_1 and Z_2.



2) Derive the transfer function T when the load resistance $R_L = \infty$.

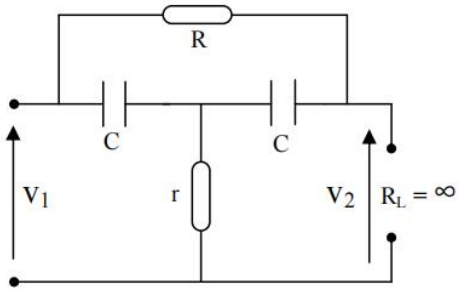
3) Sketch the Bode plot for the following cases, specifying the nature of the filter obtained:

a) Two-port network Q1: $Z_1 = R$, Z_2 is the impedance of the capacitor 2C. Determine the cutoff frequency of the filter corresponding to Q_1.

b) Two-port network Q_2: Z_1 is the impedance of the capacitor C, $Z_2 = R/2$. Determine the cutoff frequency of the filter corresponding to Q_2.

c) Two-port network Q consisting of Q_1 in parallel with Q_2. Determine the cutoff frequencies and the quality factor of the filter corresponding to Q.

Exercise 07 (Week2)

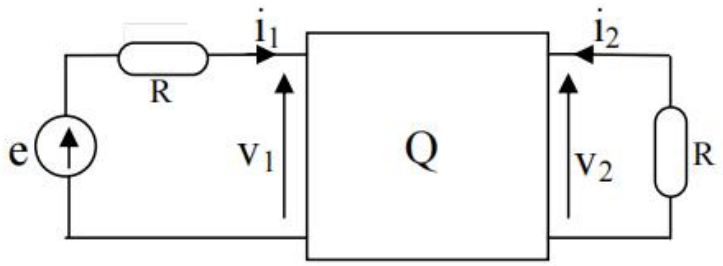


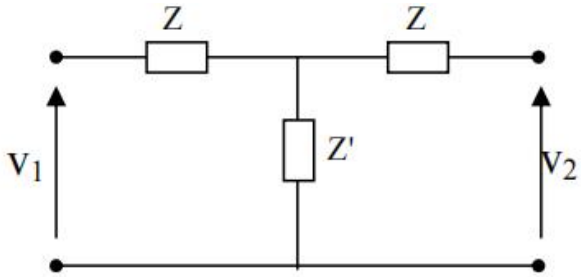
Given: The circuit with resistors R, r and capacitors C as shown in the diagram.

1) Calculate the transfer function of the "Bridged-T" two-port network shown in the figure.
2) Sketch the Bode plot. Discuss the results based on the value of the ratio R/r.

Exercise 08

Consider a two-port network (quadripole) Q, with impedance matrix elements Z_{ij}, powered by a generator (e, R) and loaded by a resistance R.





1. Express the voltage gain $G = V_2/e$ in terms of Z_{ij} and R.Assuming the network has a symmetrical T-structure, ex

2. press the matrix elements Z_{ij} in terms of Z and Z'. From this, derive the expression for the gain G as a function of R, Z, and Z'.

3. Given $Z = jL\omega$ and $Z' = R$, show that the gain G can be written in the following form:

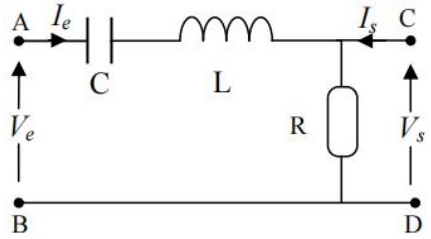
$$G = G_0 \frac{1}{(1 + j\,\omega/\omega_0)(1 + j\,\omega/3\omega_0)}$$

Define G_0 and ω_0 in terms of the circuit components.

4. Graphically represent the normalized gain magnitude expressed in dB. Determine the -3 dB bandwidth.

Exercise 09

1. Determine the transfer function T(ω) for an RLC circuit, where the output is taken across the resistor (unloaded output, meaning Is = 0).

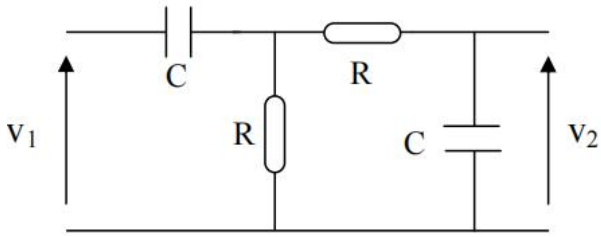


2. Express T in terms of ω/ω_0 (where ω_0 is the resonance angular frequency) and the quality factor Q. Sketch the Bode plots for T.

3. Express the gain G_{dB} for the cases $\omega/\omega_0 >> 1$ and $\omega/\omega_0 << 1$.

4. Determine the -3 dB bandwidth as a function of Q and ω_0.

Exercise 10 (Week 3)



Consider the two-port network (quadripole) shown below.

a) Predict the asymptotic behavior of this filter.

b) Calculate the transfer function $T(j\omega) = V_2/V_1$ in terms of ω and ω_0 (where $\omega_0 = 1/RC$).

c) Show that the denominator can be written as a product of first-order functions: $\left(1 + j\frac{\omega}{\omega_1}\right)\left(1 + j\frac{\omega}{\omega_2}\right)$, expressing ω_1 and ω_2 in terms of ω_0.

d) Establish the Bode plot.